Open Rail Data Wiki

# Signalling Nomenclature

Individual signalling elements (signals, track circuits, points, level crossings) are identified by means of standard nomenclature. The trackside equipment, which generates the data for the signalling elements, may include solid-state interlocking (SSI) outputs, relay contacts, push buttons, levers and other indications.

In SOP Tables and ECS's, the first letter of an individual function denotes the general kind of the element. Preceding letters denote the function of the element, the equipment code, which consists of one or more letters.

The general rule has its exceptions. For example, TRTS and Route Elements have their own naming schemes. And finally, the name of an element doesn't necessarily start with a letter.

# Element types

| Code | Meaning |
|---|---|
| L | Latch |
| P | Points |
| R | Route |
| S | Signal |
| T | Track Circuit or Axle Counter Section |

Train Ready To Start (TRTS) indications may be preceded by 'TRS' or 'TRTS'.

# Equipment code

The table below denotes the characters of an equipment code (from GK/RT0205 Appendix D (https://www.rssb.co.uk/standards-catalogue/CatalogueItem/GKRT0205-Iss-1)):

| Descriptive Term (preceding letters) | | Apparatus (last letter) | |
|---|---|---|---|
| A | Approach; Automatic; Relay Contact – Arm | A | Axle Counter |
| B | Block; Bolt; Relay Contact – Back | B | Block Instrument |
| C | Checking or Providing; Coding | C | Contact |
| D | Clear (green); Decoding; Relay Contact – De-energised | | |
| E | Light; Lamp; Heat (externally applied); Emergency; Earth | E | Light; Lamp; Earth |
| F | Fog; Flashing; Feed; Relay Contact – Front | F | Detonator Placers |
| f | Frequency | f | Fuse |
| G | Signal | G | Signal Apparatus |
| H | Caution (yellow) | H | Capacitor |
| HH | Preliminary Caution (double yellow) | | |
| I | AWS | I | Inductor |
| J | Time (delayed action) | J | Rectifier; Diode |
| K | Indicating or Detecting | K | Indicator Electro-mechanical |
| L | Locking; Left | L | Lock |
| M | Marker; Magnetic | M | Motor |
| N | Normal | N | Release; Hand Operated Switch; Push Button or Key |
| O | Retarding | O | Resistor; Heater |
| P | Repeating | P | Lever Latch or Trigger Contact |
| Q | Treadle or Bar | Q | Local Coil of a Double Element Relay; Treadle or Bar |
| R | Reverse; Right; Danger (red) | R | Relay or Contactor |
| Rx | Receiving | Rx | Receiver |
| S | Stick | | |
| T | Track Circuit | T | Transformer |
| Tx | Transmitting | t | Terminal |
| U | Route | U | Unit |
| V | Trainstop (including TPWS) | V | Trainstop (including TPWS) Apparatus |
| W | Points | W | Point Operating Apparatus |
| X | Audible Annunciator; Level Crossing; Wrong Direction | X | Audible Annunciator (bell, buzzer, horn, etc.) |
| Y | Slotting or Disengaging | Y | Slotting or Disengaging Apparatus |
| Z | Special | Z | Special Unit |

# Routes

Typically, the name of an individual route consists of three parts: the signal number, route letter and class. However, the order and number of these parts may vary.

Examples on how to interpret different naming conventions:

| Name | Signal Number | Route Letter | Class |
|---|---|---|---|
| R101(M)A | 101 | A | M |
| R101A(M) | 101 | A | M |
| R101AM | 101 | A | M |
| R101M | 101 | | M |
| RAW149BMC | AW149 | B | M and C |
| R229XA(S) | 229X | A | S |

# Route letter

The route letter ('A', 'B', 'C', ...) denotes the individual path taken from a signal through a set of points and/or lines. All signals have at least one route. The 'A' route is the furthest signalled route to the left, in the direction of travel. The following routes are lettered on alphabetic order from left to right.

# Route class

The last character is the route class (by specification, in brackets):

| Code | Type | Meaning |
|---|---|---|
| (M) | Main route | This is a route set from a main signal. The driver may proceed at main signal's aspect. |
| (C) | Call-on route | This is a permissive working for a main route and allows a train to enter a section already occupied by another train. The driver must proceed with caution (< 10 mph). |
| (S) | Shunt route | This is a route set from a shunting signal, which may be attached to a main signal or placed on the ground. The driver must pass the signal with caution at a speed which allows the train to stop short of any obstruction. |
| (W) | Warner route | This may be used when a full overlap is not available beyond the exit signal of a route. The signal will stay at danger (red) until it changes to caution (yellow) at the time the train stops in front of the signal. |
| (PS) | Proceed on Sight Authority | The signal will notionally be used where the route setting and locking function is still proved to be operable but a function such as train detection or lamp proving of a signal ahead may be failed. |

# Signals

| Code | Meaning |
|---|---|
| BR, R | Repeater signal |
| RGE | Electric lamp illuminating danger aspect (Red) |
| HGE | Electric lamp illuminating caution aspect (Yellow) |
| DGE | Electric lamp illuminating clear aspect (Green) |
| RGK | Electro-Mechanical Indicator showing signal at danger aspect (red) |
| HGK | Electro-Mechanical Indicator showing signal at caution aspect (yellow) |
| DGK | Electro-Mechanical Indicator showing signal at clear aspect (green) |
| OFF, OFFG, OFFK | Signal at any other state than danger (red) |
| LOSECK | Limit of Shunt lamp checking indicator |

# Points

| Code | Meaning |
|---|---|
| GF | Ground frame |
| FK | Electro-mechanical indicator showing ground frame points in local operation |
| N | Normal position |
| NK | Electro-mechanical indicator showing normal position |
| NWK | Electro-mechanical indicator showing points in normal position |
| R | Reverse position |
| RK | Electro-mechanical indicator showing reverse position |
| RWK | Electro-mechanical indicator showing points in reverse position |